

Geometric Series and Convergence

Section 4.3 — Howell & Mathews

Geometric Series

Definition

For a complex number $z \in \mathbb{C}$, the *geometric series* is

$$\sum_{n=0}^{\infty} z^n.$$

Theorem (Geometric Series)

$$\sum_{n=0}^{\infty} z^n = \begin{cases} \frac{1}{1-z}, & |z| < 1, \\ \text{diverges}, & |z| \geq 1. \end{cases}$$

Sequences in $\mathbb{C}$

Definition (Convergence of a Sequence)

A sequence $\{z_n\} \subset \mathbb{C}$ converges to z_0 if

$$\forall \varepsilon > 0, \exists N \text{ such that } n > N \implies |z_n - z_0| < \varepsilon.$$

Theorem

If a complex series $\sum z_n$ converges, then

$$\lim_{n \rightarrow \infty} z_n = 0.$$

Infinite Series

Definition

Given a sequence $\{z_k\}$, define the partial sums

$$S_N = \sum_{k=0}^N z_k.$$

The series $\sum_{k=0}^{\infty} z_k$ converges if $\{S_N\}$ converges in $\mathbb{C}$.

Absolute Convergence

Definition

A complex series $\sum z_n$ converges *absolutely* if the real series

$$\sum |z_n|$$

converges.

Theorem

Absolute convergence implies convergence.

Comparison Test

Theorem (Comparison Test)

Suppose $|z_n| \leq M_n$ for all n , where $\{M_n\}$ is a sequence of nonnegative real numbers. If

$$\sum M_n$$

converges, then the complex series $\sum z_n$ converges.

Cauchy Criterion

Definition

A sequence $\{z_n\}$ is a *Cauchy sequence* if

$$\forall \varepsilon > 0, \exists N \text{ such that } m, n > N \implies |z_m - z_n| < \varepsilon.$$

Theorem

Every Cauchy sequence in $\mathbb{C}$ converges.

Divergence Test

Theorem (Necessary Condition for Convergence)

If

$$\lim_{n \rightarrow \infty} z_n \neq 0,$$

then the series $\sum z_n$ diverges.

Examples: $\sum z^n$

- If $|z| < 1$, then $\sum z^n$ converges to $(1 - z)^{-1}$.
- If $|z| > 1$, then $|z^n| \rightarrow \infty$ and the series diverges.
- If $|z| = 1$, the terms do not tend to zero in general.

Summary

- Geometric series converge exactly when $|z| < 1$.
- Absolute convergence guarantees convergence.
- Comparison and Cauchy tests extend naturally to $\mathbb{C}$.
- All convergence arguments rely on the modulus.